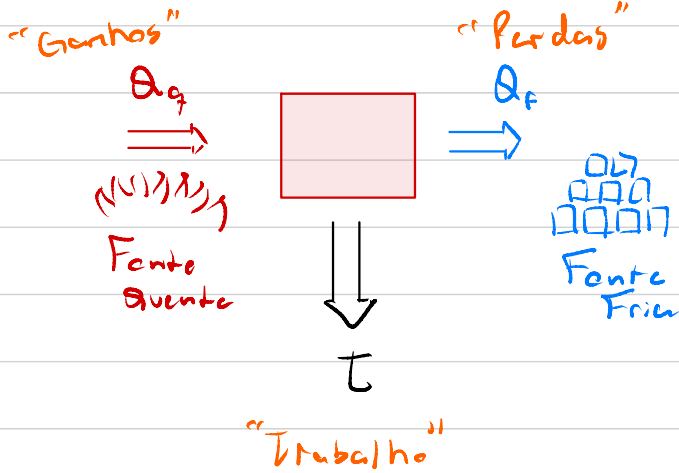


Máquinas Térmicas

Segunda Lei da Termodinâmica



Rendimento

$$\frac{Q_f}{Q_g}$$

$$i) \eta = (100\%) - (\% \text{ perdida})$$

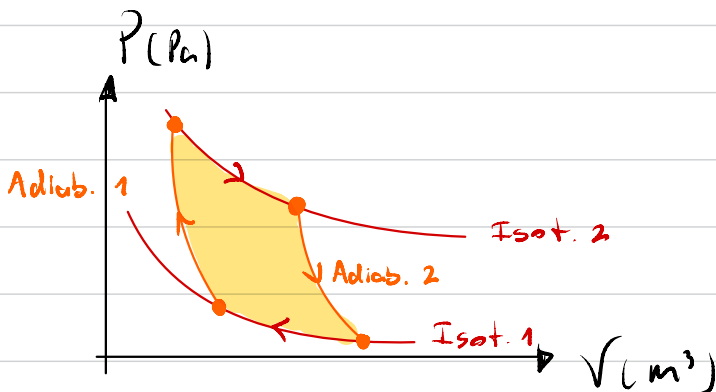
$$\eta = 1 - \frac{Q_f}{Q_g}$$

$$ii) \eta = (\% \text{ útil})$$

$$\eta = \frac{W}{Q_g}$$

Ciclo de Carnot

Rendimento Máximo



$$\eta = 1 - \frac{Q_f \rightarrow T_f}{Q_g \rightarrow T_g}$$

$\Downarrow$

$$\eta_{\text{carnot}} = 1 - \frac{T_f}{T_g}$$

$$\# \text{ Se } \eta = 100\% = 1$$

$$1 - \frac{T_f}{T_g} = 1 \quad \left\{ \begin{array}{l} \text{É impossível!} \end{array} \right.$$

* Obs: [Temp] = Kelvin!

$$T_K = T_C + 273$$

Refrigerador

